BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

Gross anatomy and neuronal integration

The written notes for the gross anatomy and neuronal integration lecture: nuclei, ganglia, tracts, and nerves, gray and white matter, the reflex arc and the types of reflexes, and neuronal circuits and pathways. Module 5 competencies 179 to 183, with eight worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 179. CNS and PNS tissue terms

In your course notes: Module 5 packet, Chapter 8, competency 179 (page 64).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Tracts and columns explained, and Afferent and efferent signals.

You should be able to: Define nucleus, ganglion, tract, and nerve and state whether each names a cluster of cell bodies or a bundle of axons in the CNS or PNS.

Four terms name groups of neurons. Two of them name clusters of neuron cell bodies, and two name bundles of axons. Which term you use depends on whether the group sits inside the CNS (the brain and spinal cord) or out in the PNS (everything else). So every one of these words answers two questions at once: cell bodies or axons, and inside or outside.

The four terms

Nucleus

A cluster of neuron cell bodies inside the CNS. The cranial nerve nuclei in the brainstem and the basal nuclei deep in the cerebrum are examples. Nuclei are made of gray matter. This meaning is different from the nucleus inside a single cell, which holds the DNA.

Ganglion (plural, ganglia)

A cluster of neuron cell bodies in the PNS. The dorsal root ganglion holds the cell bodies of sensory neurons, and the sympathetic trunk ganglia hold the cell bodies of autonomic neurons. Satellite cells surround the cell bodies in a ganglion. The basal nuclei of the brain are also called the basal ganglia, an older name that breaks this rule, since they sit in the CNS.

Tract

A bundle of axons inside the CNS. The corticospinal tract and the spinothalamic tract are examples. Tracts make up the white matter of the brain and spinal cord.

Nerve

A bundle of axons in the PNS, held together by connective tissue and supplied by blood vessels. The sciatic nerve and the vagus nerve are examples. A nerve holds axons but no cell bodies. The cell bodies of those axons sit in the CNS or in ganglia.

The same structure under two names

Structure	In the CNS	In the PNS
A cluster of cell bodies	Nucleus	Ganglion
A bundle of axons	Tract	Nerve

Term	Cell bodies or axons	CNS or PNS	Example
Nucleus	Cluster of cell bodies	CNS	A cranial nerve nucleus in the brainstem
Ganglion	Cluster of cell bodies	PNS	A dorsal root ganglion
Tract	Bundle of axons	CNS	The corticospinal tract
Nerve	Bundle of axons	PNS	The sciatic nerve

The wrappings of a nerve

On a nerve cross section, work from one axon outward. Each axon with its Schwann cell is wrapped in endoneurium. Groups of axons are bundled into a fascicle by perineurium. The whole nerve, with all its fascicles and blood vessels, is wrapped in epineurium.

Structure	What it wraps
Endoneurium	A single axon and its Schwann cell
Perineurium	One fascicle, a bundle of axons
Fascicle	A bundle of axons inside the nerve
Epineurium	The whole nerve

- Memory tip: decide first whether you are looking at cell bodies or axons. Then decide whether you are inside the CNS or outside it. Cell bodies: nucleus inside, ganglion outside. Axons: tract inside, nerve outside.
- Nucleus and ganglion are both clusters of cell bodies. The only difference between them is CNS or PNS.
- A bundle of axons inside the spinal cord is a tract, not a nerve. Nerves exist only outside the CNS. On a drawing of the cord with a nerve leaving it, the line between CNS and PNS falls at the surface of the cord where the nerve leaves, and the ganglion sits as a swelling along that nerve, outside the CNS.

Competency 180. Gray and white matter

In your course notes: Module 5 packet, Chapter 8, competency 180 (page 65).

Dr. Rennie's video for this part: Spinal cord, nerves and PNS concept video, chapter Tracts and columns explained, and Afferent and efferent signals.

You should be able to: Distinguish gray matter from white matter by composition and locate each in the brain and the spinal cord.

Nervous tissue in the CNS sorts into two kinds by its look. Gray matter is made of neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia. It is where signals are received and processed. White matter is made mostly of myelinated axons, running in bundles (the tracts), with the glia that myelinate them. It carries signals from one region to another. White matter looks white because myelin is rich in lipid.

Type	What it is made of	Where it is
Gray matter	Neuron cell bodies, dendrites, unmyelinated axons, axon terminals, and neuroglia	The cerebral and cerebellar cortex and the deep nuclei of the brain; the H-shaped inner core of the spinal cord
White matter	Myelinated axons in bundles (tracts), with their glia	Beneath the cortex of the cerebrum and cerebellum (including the arbor vitae); the outer columns of the spinal cord

In the spinal cord

In the spinal cord, gray matter is on the inside and white matter is on the outside. On a cross section, the gray matter forms a shape like a butterfly or the letter H. Its parts are the posterior (dorsal) horns, the anterior (ventral) horns, and, in the thoracic and upper lumbar cord (T1 to L2), the lateral horns. The central canal runs through the middle of the gray matter. The white matter around it is divided into posterior, lateral, and anterior columns, and these columns hold the tracts.

In the brain

In the cerebrum, the arrangement is reversed. A thin outer layer of gray matter, the cerebral cortex, covers the surface, and its large neurons include the pyramidal cells. Beneath the cortex is a large mass of white matter, including the corpus callosum. Deep inside the brain, surrounded by white matter, sit islands of gray matter called deep nuclei, such as the basal nuclei of the cerebrum and the thalamus of the diencephalon.

The cerebellum has the same plan. Its outer layer, the cerebellar cortex, is gray matter and contains the Purkinje cells. Its inner white matter branches like a tree and is called the arbor vitae. In the brainstem, nuclei of gray matter are scattered among the white matter tracts.

Region	Gray matter	White matter
Spinal cord	Inside: posterior, anterior, and lateral horns around the central canal	Outside: posterior, lateral, and anterior columns
Cerebrum	Outside: the cerebral cortex; plus deep nuclei inside	Inside, beneath the cortex
Cerebellum	Outside: the cerebellar cortex	Inside: the arbor vitae
Brainstem	Nuclei scattered inside	Tracts around and between the nuclei

- Memory tip: gray is where the cell bodies are, white is where the myelin is. The two swap places between the spinal cord and the brain, so name the organ before you name the layer.
- On the models, the cord section shows a gray butterfly in the center with white matter around it, and the cerebrum slice shows a gray cortex on the outside with white matter beneath. Your shading of the cord should be the reverse of your shading of the brain.
- If you ever write cell bodies under white matter, swap the rows. White matter is white because of myelin, and myelin is on axons.

Competency 181. Reflex arc components

In your course notes: Module 5 packet, Chapter 8, competency 181 (page 66).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Reflexes and the feedback loop.

You should be able to: Name the five components of a reflex arc in order from receptor to effector and trace a signal through them.

A reflex is a fast, automatic, predictable response to a stimulus. The pathway a reflex follows is the reflex arc. A reflex is fast because the arc is short and fixed. When the integration center is in the spinal cord, it is a spinal reflex. When it is in the brainstem, it is a cranial reflex. Either way, no conscious decision is needed, though the brain may become aware of the response afterward.

Every reflex arc has the same five components, in the same order. The signal comes in on the sensory side, is processed in the CNS, and goes out on the motor side.

The five components, in order

1. Receptor: detects the stimulus. It is the ending of a sensory neuron or a receptor cell next to one.
2. Sensory neuron: carries the signal to the CNS. In a spinal reflex, its cell body sits in a dorsal root ganglion, and its axon enters the spinal cord through the dorsal root.
3. Integration center: one or more synapses in the gray matter of the spinal cord or brainstem, where the signal passes to the next neuron. This is where the signal is processed, and it counts as one of the five components.
4. Motor neuron: carries the command away from the CNS. In a somatic spinal reflex, its cell body sits in the anterior horn, and its axon leaves through the ventral root.
5. Effector: the muscle or gland that carries out the response.

Trace a signal: the patellar reflex

The patellar (knee-jerk) reflex is the classic example. A tap on the patellar ligament, just below the kneecap, stretches the quadriceps femoris.

1. Receptor: muscle spindles in the quadriceps femoris detect the stretch.
2. Sensory neuron: carries the signal through the femoral nerve and dorsal roots into the spinal cord at L2 to L4.
3. Integration center: the sensory neuron synapses directly on a motor neuron in the anterior horn.
4. Motor neuron: carries the command out through the ventral root and the femoral nerve.
5. Effector: the quadriceps femoris contracts, and the leg extends at the knee.

Kinds of receptors

Receptors are named for the stimulus they detect.

Receptor	Stimulus it detects
Mechanoreceptors	Touch, pressure, and stretch
Thermoreceptors	Temperature
Photoreceptors	Light
Chemoreceptors	Chemicals
Nociceptors	Pain

- Memory tip: the arc runs in, through, and out: Receptor, Sensory neuron, Integration center, Motor neuron, Effector. The sensory neuron enters by the dorsal root, and the motor neuron leaves by the ventral root.
- The sensory neuron comes before the integration center and the motor neuron comes after it. If your chain has only four boxes, you left out the synapse in the cord.
- A clinician who taps below the kneecap is testing the whole arc at once. A missing reflex points to a break somewhere along that pathway: the receptor, the sensory neuron, the segment of cord, the motor neuron, or the muscle.

Competency 182. Types of reflexes

In your course notes: Module 5 packet, Chapter 8, competency 182 (page 67).

Dr. Rennie's video for this part: Nervous tissue concept video, chapter Reflexes and the feedback loop.

You should be able to: Compare somatic versus autonomic, monosynaptic versus polysynaptic, and innate versus learned reflexes and give an example of each.

Reflexes are sorted in three ways: by the effector, by the number of synapses in the integration center, and by whether the reflex is inborn or learned. Each pair answers a different question, so one reflex fits into all three. The

patellar reflex, for example, is somatic, monosynaptic, and innate.

Somatic versus autonomic: which effector?

A somatic reflex has skeletal muscle as its effector. The patellar reflex is a somatic reflex. An autonomic reflex, also called a visceral reflex, has smooth muscle, cardiac muscle, or a gland as its effector. In the pupillary light reflex, bright light makes the pupils constrict. The effector is the smooth muscle of the iris, and the integration center is in the midbrain.

Monosynaptic versus polysynaptic: how many synapses?

In a monosynaptic reflex, the sensory neuron synapses directly on the motor neuron, so there is only one synapse in the integration center. The stretch reflex, such as the patellar reflex, is monosynaptic. In a polysynaptic reflex, one or more interneurons sit between the sensory and motor neurons, so there are two or more synapses. The withdrawal reflex, pulling your hand away from a hot stove, is polysynaptic. A monosynaptic reflex is the faster of the two, because each synapse adds a short delay.

Innate versus learned: born with it or practiced?

An innate reflex is present from birth and does not have to be learned. Blinking when something touches the eye is an innate reflex. A learned reflex, also called an acquired reflex, is built through practice and repetition until it runs automatically. An experienced driver braking when the car ahead stops suddenly, or catching a suddenly dropped object, are learned reflexes.

Reflex type	Basis of the name	Example
Somatic reflex	The effector is skeletal muscle	The patellar, or knee-jerk, reflex
Autonomic reflex	The effector is smooth muscle, cardiac muscle, or a gland	The pupillary light reflex
Monosynaptic reflex	Just one synapse in the integration center	The stretch reflex, such as the patellar reflex
Polysynaptic reflex	Two or more synapses, with interneurons	The withdrawal reflex
Innate reflex	Present from birth, not learned	The blink reflex
Learned reflex	Built through practice and repetition	Braking when the car ahead stops; catching a dropped object

- Memory tip: each pair asks one question. Somatic or autonomic: which effector? Mono or poly: how many synapses? Innate or learned: born with it or practiced?
- The withdrawal reflex runs through interneurons, so it is polysynaptic, not monosynaptic.
- The pupillary light reflex is autonomic because its effector is smooth muscle in the iris, not skeletal muscle. Do not file it under somatic.

Competency 183. Neuronal pools and pathways

In your course notes: Module 5 packet, Chapter 8, competency 183 (page 68).

You should be able to: Describe convergence and divergence circuits and distinguish ascending pathways, descending pathways, and decussation.

Neurons in the CNS work in groups. A neuronal pool is a group of interconnected neurons that process the same kind of information together. A pool can hold thousands of neurons. The pattern of connections inside and between pools is called a neural circuit, and the pattern shapes how a signal is handled.

Convergence and divergence

In a convergence circuit, several presynaptic neurons synapse on one postsynaptic neuron. The one neuron can then combine input from many sources. A motor neuron in the anterior horn of the spinal cord is an example. It receives input from several regions of the brain and from sensory neurons, and its single output goes to a skeletal muscle.

In a divergence circuit, one presynaptic neuron branches and synapses on many postsynaptic neurons. One signal spreads out to reach many cells. A single motor neuron in

the brain, for example, can stimulate many motor neurons in the spinal cord, which in turn reach many muscle fibers.

Signals can also travel in series or in parallel. In serial processing, the signal passes along a single chain of neurons, as in a reflex arc. In parallel processing, the same signal travels along several pathways at the same time.

Structure	What it is
Neuronal pool	A functional group of interconnected neurons in the CNS that processes information together
Convergence	Many neurons feed their signals onto one neuron; the signal travels in toward one neuron
Divergence	One neuron spreads its signal to many neurons; the signal travels out to many neurons
Serial processing	A signal passing along a single chain of neurons, as in a reflex arc
Parallel processing	A signal sent along several pathways at the same time
Integration	The combining of all the incoming signals so that a neuron produces a single response

Ascending pathways, descending pathways, and decussation

Long-distance signals travel through the CNS in tracts that run in the white matter. A tract's name usually tells you where it starts and where it ends. The spinothalamic tract runs from the spinal cord to the thalamus.

Ascending pathways carry sensory information up the spinal cord to the brain. The spinothalamic tract, which carries pain and temperature, is an ascending pathway.

Descending pathways carry motor commands down from the brain to the spinal cord. The corticospinal tract, which runs from the cerebral cortex to the spinal cord and controls voluntary movement, is a descending pathway.

Decussation is the crossing of a pathway from one side of the CNS to the other. The main sensory and motor pathways cross, so each side of the brain senses and controls the opposite side of the body. Most corticospinal axons cross in the medulla oblongata, at the pyramidal decussation. Spinothalamic axons cross in the spinal cord, close to where the sensory fibers enter.

Structure	What it is	Which way the signal travels	Example
Ascending pathways	Tracts that carry sensory information up the spinal cord to the brain	Up the cord to the brain	The spinothalamic tract
Descending pathways	Tracts that carry motor commands down from the brain to the spinal cord	Down from the brain to the body	The corticospinal tract
Decussation	The crossing of a pathway to the opposite side of the CNS	Across the midline	The pyramidal decussation in the medulla oblongata

- Memory tip: converge means come together onto one. Diverge means spread out from one. Ascending carries sensation up, descending carries commands down, and decussation is the crossover.
- On a drawing, convergence arrows point in toward one neuron and divergence arrows spread out from one. If both circuits look the same, one is wrong.
- Because the pathways cross, the left side of the brain controls the right side of the body. On a body outline, the half the left hemisphere controls is the right half.

What this means for patients

- A tap below the kneecap triggers the patellar reflex, and a clinician watches the leg kick. That one test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the segment of spinal cord in between. A missing reflex points to a break somewhere along that pathway.
- Because the main sensory and motor pathways decussate, the right side of the brain senses and controls the left side of the body. This is why a stroke in one cerebral hemisphere causes weakness on the opposite side of the body.

- On an MRI scan, gray matter and white matter show up as different shades because myelin is rich in lipid. That contrast lets a radiologist see the cortex, the deep nuclei, and the white matter tracts, and spot damage such as the patches of lost myelin in multiple sclerosis.

Worked questions

Eight questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 179

A cluster of neuron cell bodies located in the PNS is called a

A. nucleus

B. tract

C. ganglion

D. nerve

Answer: C. A ganglion is a cluster of cell bodies outside the CNS. The dorsal root ganglion, which holds the cell bodies of sensory neurons, is the example you will see most. Satellite cells surround the cell bodies inside it.

Why the others are wrong

- **A** A nucleus is also a cluster of cell bodies, but inside the CNS, such as a cranial nerve nucleus in the brainstem.
- **B** A tract is a bundle of axons inside the CNS. It holds axons, not cell bodies.
- **D** A nerve is a bundle of axons in the PNS. It holds no cell bodies at all; those sit in the CNS or in a ganglion.

QUESTION 2 · COMPETENCY 179

The corticospinal tract is best described as

A. a bundle of axons in the CNS

B. a bundle of axons in the PNS

C. a cluster of cell bodies in the CNS

D. a cluster of cell bodies in the PNS

Answer: A. Any structure called a tract is a bundle of axons running inside the brain or spinal cord. Tracts make up the white matter of the CNS. The corticospinal tract runs from the cerebral cortex down to the spinal cord.

Why the others are wrong

- **B** A bundle of axons in the PNS is a nerve, such as the sciatic nerve. Nerves exist only outside the CNS.
- **C** A cluster of cell bodies in the CNS is a nucleus, such as the basal nuclei. A tract carries axons, not cell bodies.
- **D** A cluster of cell bodies in the PNS is a ganglion, such as the dorsal root ganglion.

QUESTION 3 · COMPETENCY 180

On a cross section of the spinal cord, where is the gray matter?

A. In the outer columns, surrounding the white matter

B. In a thin cortex on the surface of the cord

C. In the arbor vitae, branching through the center

D. In the inner H-shaped core, around the central canal

Answer: D. In the spinal cord, gray matter is on the inside. It forms a butterfly or H shape made of the posterior, anterior, and (in T1 to L2) lateral horns, with the central canal running through its middle. White matter surrounds it in posterior, lateral, and anterior columns.

Why the others are wrong

- **A** The outer columns of the cord are white matter. They hold the myelinated tracts that run up and down the cord.
- **B** A thin gray cortex on the surface is the arrangement in the cerebrum and cerebellum, not in the cord. The cord has no cortex; its gray matter is inside.
- **C** The arbor vitae is the branching white matter of the cerebellum. It is not in the spinal cord, and it is white matter, not gray.

QUESTION 4 · COMPETENCY 181

In a reflex arc, which component comes immediately after the sensory neuron?

A. Receptor

B. Integration center

C. Motor neuron

D. Effector

Answer: B. The arc runs receptor, sensory neuron, integration center, motor neuron, effector. The sensory neuron carries the signal into the CNS, where it reaches one or more synapses in the gray matter. Those synapses are the integration center, and they count as one of the five components.

Why the others are wrong

- **A** The receptor comes before the sensory neuron, not after it. It detects the stimulus and starts the signal.
- **C** The motor neuron comes after the integration center. The sensory neuron must synapse in the CNS before the command can go out on a motor neuron.
- **D** The effector is the last component, the muscle or gland that responds. Two components sit between it and the sensory neuron.

QUESTION 5 · COMPETENCY 181

In the patellar reflex, the axon of the sensory neuron enters the spinal cord through the

A. dorsal root

B. ventral root

C. anterior horn

D. central canal

Answer: A. Sensory axons enter the cord through the dorsal root, and their cell bodies sit in the dorsal root ganglion along the way. In the patellar reflex they arrive at L2 to L4 and synapse directly on a motor neuron in the anterior horn.

Why the others are wrong

- **B** The ventral root is the way out, not the way in. The motor neuron's axon leaves the cord through the ventral root to reach the quadriceps femoris.
- **C** The anterior horn is gray matter inside the cord where the motor neuron's cell body sits. It is where the sensory neuron ends up synapsing, not the route it uses to enter the cord.
- **D** The central canal is the fluid-filled channel running through the middle of the gray matter. No axons enter the cord through it.

QUESTION 6 · COMPETENCY 182

The pupillary light reflex is classified as an autonomic reflex because

A. it has only one synapse in its integration center

B. it is present from birth

C. its effector is the smooth muscle of the iris

D. its integration center is in the midbrain

Answer: C. Somatic versus autonomic is decided by the effector. Skeletal muscle makes a reflex somatic; smooth muscle, cardiac muscle, or a gland makes it autonomic. In the pupillary light reflex the smooth muscle of the iris constricts the pupil, so the reflex is autonomic.

Why the others are wrong

- **A** The number of synapses decides monosynaptic versus polysynaptic, not somatic versus autonomic.
- **B** Being present from birth makes a reflex innate. That is the innate versus learned pair, a separate question from the effector.
- **D** The pupillary light reflex does have its integration center in the midbrain, which makes it a cranial reflex, but the location of the integration center is not what makes a reflex autonomic. The effector is.

QUESTION 7 · COMPETENCY 182

A reflex in which one or more interneurons sit between the sensory neuron and the motor neuron is called

A. monosynaptic

B. polysynaptic

C. somatic

D. innate

Answer: B. Interneurons add synapses. With one or more interneurons in the chain there are two or more synapses in the integration center, which is the definition of a polysynaptic reflex. The withdrawal reflex, pulling your hand off a hot stove, is the example. Each extra synapse adds a short delay, so a polysynaptic reflex is a little slower than a monosynaptic one.

Why the others are wrong

- **A** In a monosynaptic reflex the sensory neuron synapses directly on the motor neuron, with no interneuron and only one synapse. The stretch reflex, such as the patellar reflex, is monosynaptic.
- **C** Somatic describes the effector, skeletal muscle. A somatic reflex can be monosynaptic (patellar) or polysynaptic (withdrawal), so the word says nothing about interneurons.
- **D** Innate means present from birth. It is about whether the reflex was learned, not about how many synapses it has.

QUESTION 8 · COMPETENCY 183

A stroke in the left cerebral hemisphere causes weakness on the right side of the body because

A. ascending pathways carry motor commands to the right side

B. the left hemisphere controls both sides of the body

C. the spinothalamic tract carries motor commands to the right side

D. corticospinal axons cross to the opposite side at the pyramidal decussation

Answer: D. The corticospinal tract is the descending motor pathway for voluntary movement. Most of its axons decussate, or cross, in the medulla oblongata at the pyramidal decussation. So the left cortex commands the muscles of the right side, and damage on the left shows up as weakness on the right.

Why the others are wrong

- **A** Ascending pathways carry sensory information up to the brain. Motor commands travel down descending pathways.
- **B** Each hemisphere controls the opposite side of the body, not both sides. If it controlled both, a one-sided stroke would not produce one-sided weakness.
- **C** The spinothalamic tract is an ascending sensory pathway for pain and temperature. It does cross in the cord, but it carries sensation up, not commands down.